

Math 306 Group Quiz 3

Names: _____

1. Define $T : M_{2,2}(\mathbb{R}) \rightarrow M_{2,2}(\mathbb{R})$ by $TA = A + A^\top$.

a. Find all eigenvalues/vectors of T .

b. Is T diagonalizable? If so, what is the matrix $M(T)$ in the basis of eigenvectors you found above?

2. Let $T : V \rightarrow V$ be linear and let U be a subspace of V . Define U to be T invariant if $Tu \in U$ for all $u \in U$. Prove that these are T invariant subspaces:

a. $\text{range } T$

b. $\text{null } T$

c. $\{v \in V : Tv = \lambda v\}$ for some fixed $\lambda \in \mathbb{F}$.

d. $\text{span}\{v, Tv, T^2v, \dots\}$ for some fixed $v \in V$.